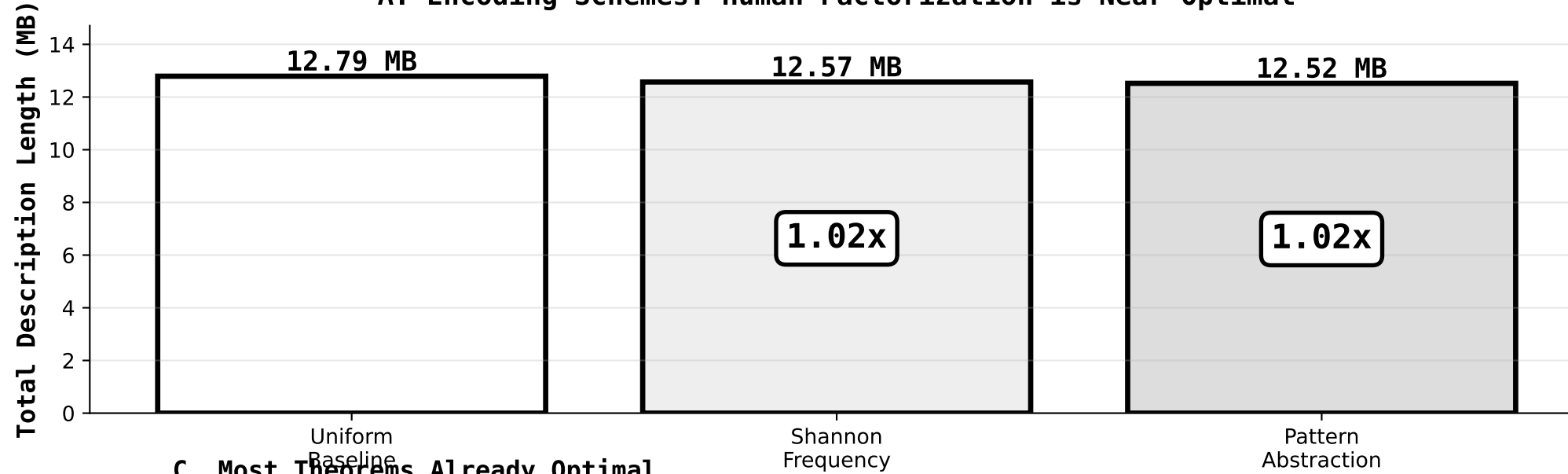


Mathlib Description Length: Comprehensive Summary

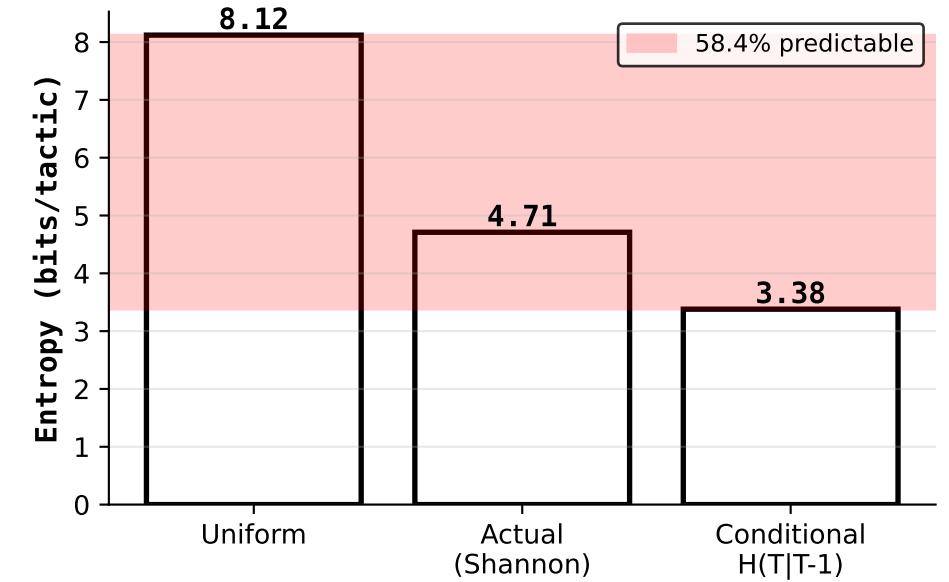
Measuring Information-Theoretic Compression of Formal Mathematics

A. Encoding Schemes: Human Factorization is Near-Optimal



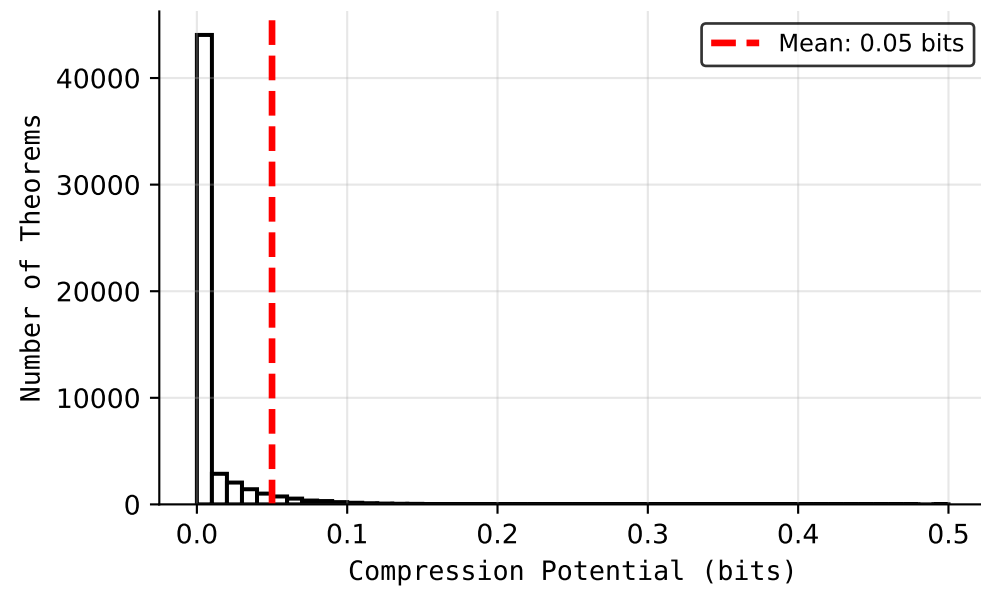
B. Tactic Predictability

58.4% Reduction

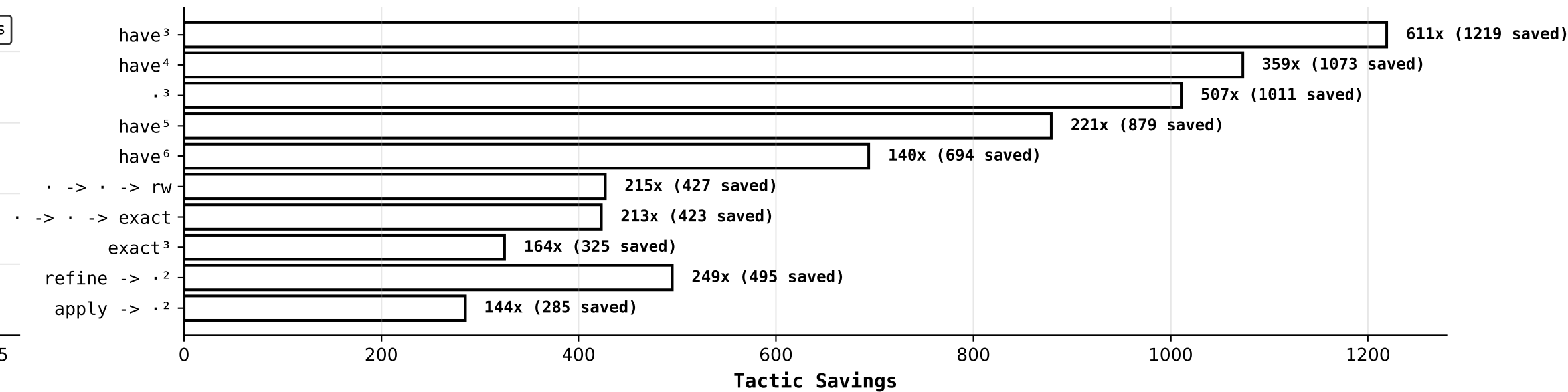


C. Most Theorems Already Optimal

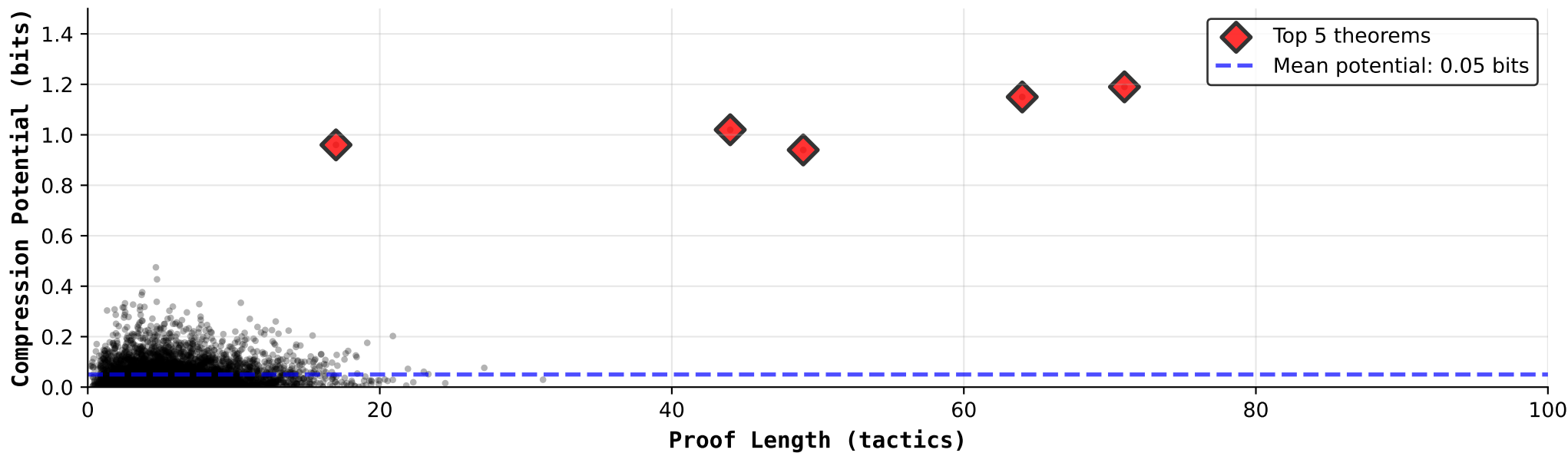
Avg: 0.05 bits



D. Top 10 Crystallization Candidates: Repetitive Patterns Dominate



E. Compression Landscape: Long Proofs ≠ High Compression



F. Key Findings

SUMMARY STATISTICS	
Corpus Size:	
Total theorems:	126,792
Tactic proofs:	54,477 (43%)
Term proofs:	72,315 (57%)
Vocabulary:	
Unique tactics:	278
Unique premises:	70,863
Description Length:	
Uniform encoding:	12.79 MB
Shannon encoding:	12.57 MB
Pattern-optimal:	12.52 MB
Compression:	
Frequency gain:	1.7%
Pattern gain:	0.4%
Total headroom:	2.1%
Crystallization:	
Valuable patterns:	9,068
Tactic savings:	81,727
FINDING: Human factorization is information-theoretically optimal. Only 2.1% compression headroom (not the predicted 36%).	